

Simplified Mathematical Representations of Heavy-Duty Gas Turbines

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This paper provides simplified mathematical representations of heavy-duty gas turbines that are suitable for use in dynamic power system studies and in dynamic analyses of connected equipment. The full range of heavy-duty, single-shaft gas turbines, from 18 MW (24,000 HP) and 106 MW (142,000 HP), is covered, as well as both liquid and gas fuel systems, parallel and isolated operation, droop and isochronous governors, and the impact of both air- and hydrogen-cooled generators on the characteristics of the model. Alterations to the models, for the purpose of further simplification, are also discussed.

Introduction

During the past several years, requests have been received for simplified mathematical models of General Electric gas turbine generators to be used as input for comprehensive studies of gas turbine users' power systems. The intent of these studies has been the investigation of power system stability, the development of dispatching strategy, and contingency planning for system upsets. Invariably, each request has placed the emphasis on keeping the model "simplified" yet flexible enough to represent different units with different characteristics. This paper is intended to answer these requests with a unified approach that offers the broadest possible base of applicability.

Terminology used throughout the block diagrams and supplementary tables is shown in the nomenclature section. In general, this terminology is kept in the per unit system with the exception of temperatures. (Per unit values are the decimal equivalents of percent values. In the percent system, 100 equals the design value, while in the per unit system 1.00 equals the design value.) Real equivalents of per unit values for the values for the various machines sizes are presented in both English and metric units, as are equations associated with the temperature control system. Notes to the figures are kept sequentially, and a specific note such as No. 1, for Fig. 1, will also apply to Figs. 4, 5, and 6 as called out in the specific figure. The system gains, coefficients, and time constants included in this paper represent design and calculated values. They have been verified by test and actual field experience accumulated from numerous installations in many different applications.

The subject of this paper has been limited to simple cycle, single-shaft, generator drive gas turbines only, because inclusion of combined cycle installations or two-shaft gas

Table 1 Limits of applicability

1 For simple cycle, single-shaft, generator drive only
2 Allowable speed range = 95 to 107 percent of rated speed
3 Open inlet guide vanes only, i.e. no heat recovery
4 ISO conditions apply, i.e.,
Ta = 59°F Pa = 14.696 psia.
= 15°C = 101.325 kpa.

turbines would tend to make the paper inordinately long and complex. Specific limits of applicability of the information contained are shown in Table 1. Item 2, limitation of the allowable speed range to 95–107 percent of rated speed, is intended to cover the expected operating speed range in its extremes, while still preventing excursions into the regions where the gas turbine no longer responds linearly or where discontinuities due to control actions are encountered. It should also be noted that this paper concentrates on the functional representation of the gas turbine and its control, and is therefore independent of whether the control is Speedtronic² Mark I, or Mark II or Mark IV.

Simplified Single-Shaft Gas Turbine Simulation

Figure 1 represents a simplified block diagram for a single-shaft gas turbine, together with its control and fuel systems, as it would be represented for isolated generator drive service. Notes 1 through 8 specify details applicable to this model, and Table 2 specifies unit ratings and per unit values, by turbine Model Series and generator type, in both English and metric units.

The control system includes speed control, temperature control, acceleration control, and upper and lower fuel limits. The representation of the speed governor is suitable for either droop or isochronous control and operates on the speed error formed between a reference made up of one per unit speed plus the digital setpoint, compared with actual system or rotor speed. A droop governor is a straight proportional speed

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Copies will be available until January 1984.

²Registered trademark of the General Electric Company

³Notes for Fig. 1:

1 Digital setpoint time for -5 percent to +7 percent speed change

Model	Synchronizing	Time for 12% Change*		
		Manual	Normal	Fast load
5001M,P	6 Min. GES	1.5 Min.	12 Min.	1.5 Min.
6001B		1.5 Min.	12 Min.	~
7001B,E	4.5 Min. μ -Sync	18 Min.	36 Min.	4.5 Min.
9001B		12 Min.	48 Min.	~
9001E		18 Min.	36 Min.	~

*No-load to ISO-rated load is 1/3 of time shown if droop setting is 4%.

2 Speed governor transfer function coefficients

Type	w	x	y	z
Droop	K_D	0	.05	1
Isoc.	30	2.5	.10	0

$K_D = 1/\text{Droop}$; $K = 25$ typically for 4 percent droop setting.

Droop setting is adjustable from 2 to 10 percent.

3 VCE' limits are typical by Max = +1.5 PU; Min = -0.1 PU (no load = 0.0 PU; rated load = +1.0 PU)

4 Fuel system characteristics

Type	Model	a	b	c	τ_F	K_F
Gas	All	1	0.05	1	0.40	0
Liquid	5&6	10	1	0	0.10	1
Liquid	7&9	1	0.20	1	0.10	0

5

$$\begin{aligned} f_1 &= T_X = T_R - 700(1 - W_F) + 550(1 - N) \text{ in English units} \\ &= T_Y = T_R - 390(1 - W_F) + 306(1 - N) \text{ in metric units} \end{aligned}$$

$$6 \quad f_2 = \text{turb. torque} = 1.3(W_F - .23) + .5(1 - N)$$

7 Gas turbine dynamic characteristics

Model Series	ϵ_{CR}	τ_{CD}	ϵ_{TD}
5 & 6	.01	.10	.02
7 & 9	.01	.20	.04

8 $\tau_T = 450$ in $^{\circ}\text{F}$ or 250 in $^{\circ}\text{C}$

Nomenclature¹

a, b, c	= fuel system transfer function coefficients
w, x, y, z	= governor transfer function coefficients
K_D	= governor gain
K_F	= fuel system feedback

N = per unit turbine rotor speed
 s = Laplace operator
 T_R = turbine rated exhaust temperature, °F or °C
 T_X = turbine exhaust temperature, °F or °C
 VCE' = per unit fuel command/per unit speed

W_F	= per unit fuel flow
ϵ_{CR}	= combustion reaction time delay
ϵ_{TD}	= turbine and exhaust system transport delay
τ_{CD}	= compressor discharge volume time constant
τ_F	= fuel system time constant
τ_I	= turbine rotor time constant
τ_T	= temperature controller integration rate

¹Units for all time functions are in seconds except for digital set point times, which are in minutes.

Table 2 Turbine-generator characteristics and constants⁴

English units						
Model	Turbine rpm	Rating MW	Rated T_R °F	Torque Ft-Lb	Inertia Lb-Ft ²	τ_I s
5001M	5100	18.2	955	25,200	24,600	16.2
5001P	5100	24.8	919	34,200	36,100	17.5
6001B	5100	35.9	1020	49,500	44,900	15.1
7001B	3600	60.0	950	117,400	185,900	18.5
7001E	3600	75.0	1004	146,700	176,400	14.1
7001E ^a	3600	75.0	1004	146,700	153,000	12.2
9001B	3000	82.6	950	193,800	500,300	25.2
9001E	3000	106.7	972	250,400	474,700	18.5
9001E ^a	3000	106.7	972	250,400	439,500	17.1

Metric units						
Model	Turbine rpm	Rating MW	Rated T_R °C	Torque Kg-M	Inertia Kg-M ²	τ_I s
5001M	5100	18.2	513	3,484	1,037	16.2
5001P	5100	24.8	493	4,728	1,521	17.5
6001B	5100	35.9	549	6,844	1,892	15.1
7001B	3600	60.0	510	16,231	7,834	18.5
7001E	3600	75.0	540	20,282	7,434	14.1
7001E ^a	3600	75.0	540	20,282	6,447	12.2
9001B	3000	82.6	510	26,793	21,083	25.2
9001E	3000	106.7	522	34,619	20,004	18.5
9001E ^a	3000	106.7	522	34,619	18,521	17.1

ISO package ratings are shown for base load and distillate Gas ratings are normally 2 percent higher.

^aHydrogen-cooled generator; all others are air-cooled

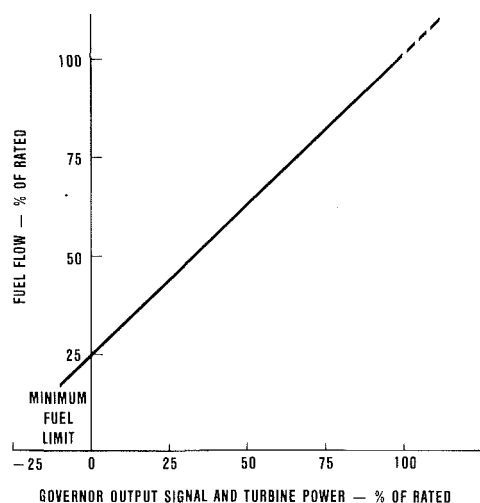


Fig. 2 Relationship between governor output signal and turbine fuel flow

with the measuring system. Under normal system conditions, where gas turbine output is determined by the slow rate of digital setpoint, these time constants are of no significance to the load limiting function. However, where increasing gas turbine output is the result of a reduction of system frequency and therefore may occur quite rapidly, exhaust temperature measurement system time constants will result in some transient overshoot in load pickup. The design of the temperature controller is intended to compensate for this transient characteristic.

Acceleration control is used primarily during gas turbine startup to limit the rate of rotor acceleration prior to reaching governor speed, thus ameliorating the thermal stresses encountered during startup. This control serves a secondary function during normal operation, in that it acts to reduce fuel flow and limit the tendency to overspeed in the event that the turbine generator separates from the system. This action is most beneficial where separation is the result of a breaker

other than the generator breaker, in which case the digital setpoint is *not* reset to 100.3 percent as previously described.

These three control functions—speed governing under part-load conditions, temperature control acting as an upper limit, and acceleration control to prevent overspeeding—are all inputs to a low value selector. The output of the low value selector, which is called *VCE'*, is the lowest of the three inputs, whichever requires the least fuel. Transfer from one control to another is bumpless and without any time lags. The output of the low value selector is compared with maximum and minimum limits. Of the two, the maximum limit acts as a backup to temperature control and is not encountered in normal operation; the minimum limit is the more important dynamically. This is because the minimum limit is chosen to maintain adequate fuel flow to insure that flame is maintained within the gas turbine combustion system. This is typically set at a torque deficiency of approximately 10 percent, is a “hard” limit, and represents a maximum rate of deceleration or torque absorption from the system.

This capability of transiently absorbing power from the connected system is unique to the gas turbine and sets it apart from other large prime movers such as steam and hydro turbines. It is normally limited to a maximum steady-state deficiency of 5 percent by the reverse power relay, which will open the generator breaker on an inverse time-current characteristic. The reverse power relay setting is adjustable and in special circumstances can be set to take full advantage of the negative power capability of the gas turbine. It should be noted that the reverse power relay is not a protective device, but is used primarily as part of the normal shutdown sequencing.

Gas turbine fuel systems are designed to provide energy input to the gas turbine in proportion to the product of the command signal (*VCE'*) times the unit speed. This is analogous to the actual mode of operation of the fuel system, since liquid fuel pumps are driven at a speed proportional to turbine rotor speed and since gas fuel control is accomplished in two stages, with the output pressure of the first stage being

⁴Turbine constants are subject to change without notice

proportional to rotor speed. The net effects is to reduce the turndown ratio of the primary fuel control from approximately 50:1 to 6:1.

From a dynamic standpoint, gas turbines have many differences from steam turbines. One of the more obvious differences is the need for a significant fraction of rated fuel to support self-sustaining, no-load conditions. This amounts to approximately 23 percent and is one of the economic driving forces to minimize operating time at full-speed, no-load conditions. As a result, the active fuel control range of the governor is from 16–100 percent, thus providing for a negative torque contribution of approximately 10 percent, as previously mentioned. To allow the use of governor parameters comparable to steam and hydro turbines, the governor operates from 0–100 percent of active load range. This range must then be corrected to 23–100 percent fuel flow range to be compatible with the thermodynamics of the turbine. This relationship between governor output signal and fuel flow is shown in Fig. 2.

As mentioned previously, the fuel gas control system consists of two valves in series, the first of which controls the pressure between the two valves as a function of speed. The second valve has a linear area versus lift characteristics and is aerodynamically designed so that sonic velocities are attained at the controlling area with flange-to-flange valve pressure ratios as low as 1.25. If valve position is maintained proportional to the VCE' signal, the net result is a flow rate of fuel gas which is proportional to the produce of VCE' and turbine speed. The response of the pressure control portion of the system may be neglected for this type of simplified study

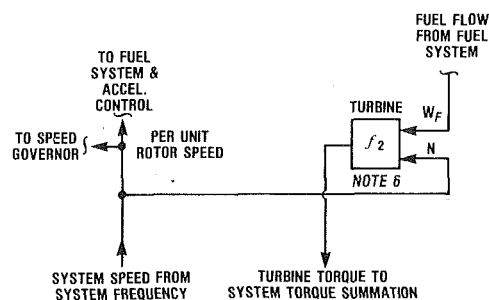


Fig. 3 Modification to Fig. 1 for parallel operation (Note 9: add turbine-generator rotor inertia to overall system inertia; Note 10: system calculations are by user)

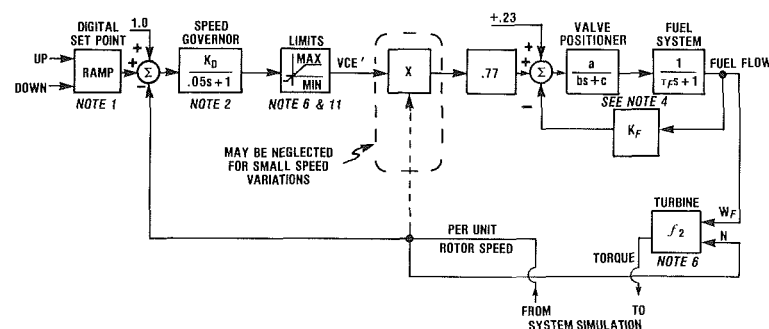


Fig. 4 Additional simplifications to block diagram of Fig. 1 (Note 11: Set Max VCE' limit = rated load = + 1.0 PU)

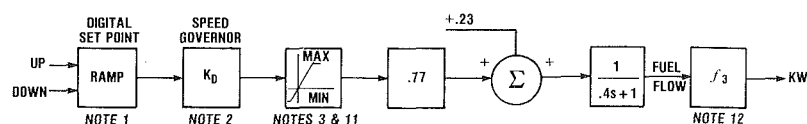


Fig. 5 Simplified block diagram for very "stiff" systems (Note 12: $f_3 = KW = 1.3(W_F - .23)$)

because of the small volume between the two valves and the higher gain of the first or pressure controlling valve. Therefore, there are only two time constants of importance in the gas system. The first is that associated with the gas control valve positioning system, and the second is the volumetric time constant associated with the downstream piping and fuel gas distribution manifold.

There are two types of liquid fuel control systems that are used on single-shaft gas turbines. Both of these are bypass systems in which part of the output of a constant volume fuel pump is returned to the pump suction. The amount of fuel bypassed is determined by measuring fuel flow going to the turbine and making that number agree with the total fuel command, which is the product of VCE' and turbine speed. The larger gas turbines utilize a closed position loop on the bypass valve, which is an inner loop within the flow feedback signal. The smaller gas turbines, the MS5001 and 6001 series, do not utilize an inner position feedback signal, and therefore close the loop around a pure integrating positioner on the bypass valve by means of only the flow feedback signal. Fuel compressibility and inertia in the system between the bypass valve and the flow measurement are represented by a relatively small time constant.

The gas turbine proper is essentially a linear, nondynamic device with the exception of the rotor time constant. There is a small transport delay associated with the combustion reaction time, a time lag associated with the compressor discharge volume, and a transport delay to transport the gas from the combustion system through the turbine. Representative values are shown in Note 7 of Fig. 1. In the context of the rotor and connected system time constants and the relatively slow setpoint ramp rates, these short-term dynamic characteristics can be ignored.

Both the torque and exhaust temperature characteristics of the single-shaft gas turbine are essentially linear with respect to fuel flow and turbine speed over the 95–107 percent speed range to which these models are limited. The torque equation is accurate to within 5 percent at part load, and is significantly more accurate at the 100 percent design rating. The exhaust temperature equations are somewhat less accurate at part load; however, since temperature control is only active at the design point, the impact of the part load inaccuracy is negligible to the overall simulation. Table 2 provides basic turbine characteristics in both English and metric units as a function of Model Series. Some of the larger gas turbines are

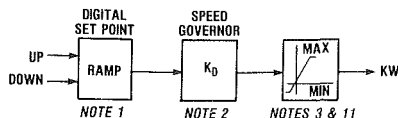


Fig. 6 Simplified block diagram for very large "stiff" systems

available with either air- or hydrogen-cooled generators. This only affects the inertia, and therefore the rotor time constant, as shown in Table 2. By way of definition, the turbine rotor time constant is defined as the time necessary for the rotor to double its speed if the initial rate of speed change is maintained after removal of rated load torque.

Simulation of Parallel Operation

The configuration shown in the lower right-hand corner of Fig. 1 is indicative of the system configuration for simulating an isolated generator drive gas turbine. Most system studies involve the simulation of numerous units all operating in parallel. With the exception of system torque balances and frequency calculations, all previous discussions apply to the models regardless of whether the gas turbine is operating as an isolated unit or in a parallel configuration. The changes necessary to merge the gas turbine model with a system model are shown in Fig. 3. In this case, the turbine generator rotor inertia should be added to the overall system inertia.

Simplifications to the Models

Most simplifications to the model presented in Fig. 1 are associated with parallel operation in a relatively "stiff" system, in which speed variation is significantly less than the 95–107 percent limitation of the model of Fig. 1. Under these conditions of limited frequency variation, several simplifications can be made to reduce the complexity of Fig. 1. First of all, since parallel operation is the only type envisioned, the speed governor can be changed to a droop-only configuration. Secondly, if the frequency variations are not greater than ± 1 percent, acceleration control will not become active except under load-loss situations, and acceleration control can be eliminated. Lastly, since gas turbine output is predominantly controlled by the digital setpoint, the need for the interactive limit imposed by temperature control is significantly diminished, thereby allowing elimination of temperature control, as well as the need for the low value

selector. Where temperature control has been eliminated, the upper load limit can be satisfactorily approximated by reducing the maximum fuel limit to level corresponding to rated load. These changes are shown in Fig. 4. Previous notes from prior figures still apply as noted. As mentioned previously, the minor dynamic characteristics of the gas turbine can also be eliminated.

Further simplification of Fig. 4 is possible in the case of parallel operation where the system inertia relative to the gas turbine is very large and system time constants are very long. For this situation many of the shorter time constants associated with the speed governor and fuel control systems can be ignored. Again, if frequency variations are small enough so as to be negligible, then fuel flow and therefore turbine output, will be a direct function of the speed governor command. These additional changes are shown in Fig. 5.

In the final analysis, where system time constants are very long relative to those associated with the gas turbine, gas turbine output will be essentially proportional to the position of the digital setpoint times the governor gain. This is shown in Fig. 6. Although this last simplification may appear to make the model trivial, it is nevertheless an adequate representation of a relatively small gas turbine in a very large system. There are no guidelines that can be put down with regard to how far model simplification should be carried since it is very much a function of the size and characteristics of the connected system. Considerable engineering judgment and detailed knowledge of equipment characteristics are required to insure against oversimplification.

Summary

This paper has attempted to present mathematical representations of General Electric heavy-duty gas turbines that would be suitable for use as input to power system stability studies. The models have been tailored to cover the full spectrum of generator drive gas turbines and appropriate turbine-generator characteristics; constants have been included in both English and metric units. All conventional fuel systems, for both liquid and gas, have been discussed, and special operating conditions such as isolated operation and use of isochronous governors have also been touched upon. Simplifications to the models, and the appropriate rationale have also been discussed. Engineering judgment must be used in conjunction with careful evaluation of the intended purpose prior to use of the simplified models.